

Yaohui Shu

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RESEARCH INTERESTS

Cloud microphysics and turbulence-microphysics interactions: stochastic condensation, broadening of cloud droplet size distributions, and entrainment-mixing. Langevin and Fokker-Planck models with finite-memory forcing, the link between maximum-entropy theory and stochastic dynamics, and large Monte Carlo ensembles with analytical benchmarks. Information-theoretic causal discovery for geophysical systems with feedback and incomplete observations.

EDUCATION

Ph.D., Applied Mathematics and Statistics 2022 – May 2027 (expected)
Stony Brook University, Stony Brook, NY

Dissertation: *Uncovering Causal Mechanisms of Aerosol-Cloud-Turbulence Interactions Using Physics- and Data-Driven Methods*

Advisors: Xiaolin Li (Stony Brook University) and Yangang Liu (Brookhaven National Laboratory)

Advanced to candidacy, Fall 2025. GPA: 3.78 / 4.0

Coursework: Advanced Stochastic Models and Risk Analysis, Differential Equations I and II, Numerical Analysis I–III, Principles of Parallel Computing, Applications of Complex Analysis, Foundations of Quantitative Finance, Financial Derivatives and Stochastic Calculus, Quantitative Risk Management.

M.S., Finance, University of Rochester, Rochester, NY. GPA: 3.81 / 4.0 Aug 2021 – Aug 2022

Certificate in Quantitative Finance (CQF), CQF Institute

RESEARCH EXPERIENCE

Doctoral Researcher Aug 2023 – Present

Department of Applied Mathematics and Statistics, Stony Brook University, with the Environmental Science and Technologies Department, Brookhaven National Laboratory

Sole Ph.D. student on the group's stochastic condensation and causality program: research direction, derivations, code, manuscripts, and weekly reports to the principal investigators and visiting scientists.

Maximum entropy and Langevin stochastic condensation

- Showed that maximum-entropy systems theory (third-order Weibull spectrum, relative dispersion 0.363) and Langevin stochastic condensation (second order, 0.523) disagree under the same conservation constraint because of the choice of size variable. Relative dispersion is the standard deviation of droplet radius divided by its mean (dimensionless).
- Generalized stochastic condensation to any conserved moment order under one hypothesis (size-independent exchange of the conserved budget), reduced three Langevin realizations to a single system, and showed that the conserved order sets the spectral shape while the scale moment sets only the scale.
- Monte Carlo runs for orders 1 to 6 (10^5 droplets) reproduce the predicted narrowing. Traced the residual offset to the small-size boundary and showed that in the finite-memory model no boundary rule can conserve probability, leave the saturation ratio untouched, and avoid accumulation at the origin at once.

Entrainment-mixing in a stochastic condensation framework

- Built a three-state model (steady adiabatic cloud, entrainment by unbiased thinning, then homogeneous or extreme-inhomogeneous mixing) and tracked the homogeneous-mixing measures of Lu et al. (2013). Both pathways reach the same liquid water content with distinct spectral evolutions, a benchmark for entrainment-mixing parameterizations.

- Developed a general scheme that moves continuously between the two mixing limits by controlling droplet number concentration and liquid water content.

Causal discovery for systems with feedback and unmeasured drivers

- Methodology targeting partial ancestral graphs over a lag window under sigma-separation, with nearest-neighbor conditional mutual information tests, an assumption ledger that resolves conflicts among component methods, and a surrogate-calibrated diagnostic separating nonlinearity from nonstationarity.
- Proposed six postulates for a time-resolved measure of causal strength and a candidate meeting five: the relative entropy rate between path measures with one drift dependence removed (closed form by Girsanov's theorem).

Research Assistant

Jan 2024 – Jan 2026

Environmental Science and Technologies Department, Brookhaven National Laboratory, Upton, NY

Supported by the DOE Atmospheric System Research (ASR) Program and BNL Laboratory Directed Research and Development (LDRD-25-008).

Stochastic condensation with information flow for causality analysis

- Generalized the McGraw-Liu (2006) Brownian drift-diffusion model, valid only for strongly damped fluctuations, to a coupled Langevin system with an Ornstein-Uhlenbeck saturation ratio spanning the full memory continuum; the original model is recovered as the decorrelation rate goes to infinity.
- Embedded the Liang-Kleeman information-flow formalism and derived the time-resolved flow from saturation ratio to droplet size in closed form, with its limits; the reverse flow is zero by construction.
- Built an ensemble estimator using covariances across droplets at each time. It matches theory, stays near zero in the reverse direction, and identifies causal direction where the symmetric correlation cannot.
- Ran Monte Carlo sweeps on HPC/SLURM (10^5 droplets, five seeds, decorrelation rate 0.1 to 10 s^{-1} , saturation standard deviation 0.001 to 0.02) with number- and mass-conserving boundary handling and convergence studies. Broadening grows with correlation time and fluctuation intensity, and causal coupling weakens as the distribution broadens. Published in [1].

PUBLICATIONS

Journal articles

- [1] **Shu, Y.**, Liu, Y., Zhang, T., McGraw, R., and Li, X., 2026: Stochastic condensation with information flow for causality analysis. *Geophysical Research Letters*, **53**(17), e2026GL123056. doi:10.1029/2026GL123056. Open access.

Manuscripts in preparation

- [2] **Shu, Y.**, Liu, Y., and Li, X.: Stochastic condensation and the maximum entropy cloud droplet spectrum: The order of the conserved moment. For *Physical Review E*.
- [3] **Shu, Y.**, Liu, Y., and Li, X.: A new stochastic condensation model for investigating turbulent entrainment-mixing processes. For *Journal of the Atmospheric Sciences*.
- [4] **Shu, Y.**: Entropy-based causal discovery for spatiotemporal dynamical systems with feedback and unmeasured drivers. Methodology working paper.

Software

- `stochastic_condensation_info_flow` (v1.2), code for [1]. MIT License. doi:10.5281/zenodo.22089240; GitHub.

CONFERENCE ABSTRACTS AND PRESENTATIONS

- Liu, Y., **Shu, Y.**, Zhang, T., and McGraw, R., 2026: A new generalized stochastic model and impacts of microphysical fluctuations. *106th AMS Annual Meeting*, January 2026 (presented by Y. Liu).
- **Shu, Y.**, Liu, Y., Zhang, T., and McGraw, R. L., 2025: Stochastic condensation/evaporation with different physical constraints and information flow for causal discovery. Poster, *AGU Fall Meeting 2025*,

A21B-2034.

- Liu, Y., Gao, S., Lu, C., and **Shu, Y.**, 2025: Relationship between turbulent entrainment-mixing and droplet clustering. *AGU Fall Meeting 2025*, A21B-203.
- Liu, Y., **Shu, Y.**, Zhang, T., and McGraw, R., 2025: Stochastic condensation with information flow for causality analysis. Poster, *DOE Atmospheric System Research (ASR) Program Meeting*, 2025.
- Group talks, Brookhaven National Laboratory and Stony Brook University: *A new generalized entrainment-mixing scheme* (February 2026); *Information flow* (July 2025).

TEACHING

Teaching Assistant, AMS 595 Fundamentals of Computing Aug 2026 – Present
Stony Brook University. Graduate course in scientific programming with MATLAB, Python, and C++.

TECHNICAL SKILLS

Programming	Python (NumPy, SciPy, pandas, TensorFlow), MATLAB, C++, R, L ^A T _E X, Git
HPC	SLURM clusters, multiprocessing, vectorized and chunked ensemble workflows
Simulation	SDE integrators, Monte Carlo convergence studies, conserving boundary schemes
Stochastic analysis	Itô calculus, Fokker-Planck/Langevin equations, OU processes, maximum entropy
Causal inference	Information flow, transfer entropy, conditional mutual information, DML/AIPW/TMLE
Machine learning	LSTM, XGBoost, PCA, self-organizing maps, hyperparameter optimization

ADDITIONAL TRAINING

Quantitative Finance Capstone, CQF Institute 2023
LSTM prediction of equity return direction from 156+ engineered features, reduced to 10–15 predictors with XGBoost importance, self-organizing maps, and PCA; 162-configuration hyperparameter search with leakage-free splits.